

A mechanistic modeling framework for CD8⁺ T-cell antigen immunogenicity

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Abstract

CD8⁺ T-cell immunogenicity depends on antigen presentation, the availability of cognate T-cell receptors after selection, and productive peptide–MHC–TCR interaction. We develop a mechanistic score that represents these requirements through the multiplicative response $F = P \cdot N \cdot Q \cdot \Pi$. Recognition geometry, selection decisions, and alternative constructions of the modeled HLA environment define candidate minimal-epitope scores. Long-peptide assays additionally require an aggregation rule over candidate epitope–HLA presentations. These choices jointly specify the composite models confronted with four refutation datasets: resolved NCI neoepitope observations, PDAC vaccine responses, and SARS-CoV-2 spike-mutation and non-spike responses. A directed supported-superiority tournament asks which models remain admissible in every context under declared prevalence, case-mix, and computational challenges. The full comparison exposes each complete model to all four datasets, but requires extensive score construction and pairwise assessment. We formulate sequential approximations that expand NCI-retained regimes into long-peptide models and distinguish restrictions on eligible candidates from restrictions on their challengers. The resulting design preserves unresolved alternatives and makes explicit which criticisms a tractable comparison can omit.

Candidate antigens vastly outnumber those that can be tested experimentally. Predictive models are therefore used to prioritize peptides for immune monitoring, vaccine design, and therapeutic development. Predictive models are constructed from observations obtained in particular hosts, diseases, exposure conditions, and assays. The models are then used to prioritize candidate antigens under different combinations of those circumstances.

Many approaches address this problem by associating observed immunogenicity labels with molecular descriptions of the interacting participants. These descriptions may include peptide sequence and structure, MHC sequence and structure, and, when available, TCR sequence or structural information. Increasing the richness of this representation can better characterize molecular compatibility. It does not, however, specify whether the represented interaction is realized within the biological circumstances required to produce a response.

CD8⁺ T-cell immunogenicity is not an intrinsic property of either an antigen or a peptide–MHC–TCR configuration described by sequence and structure alone; molecular compatibility is therefore insufficient to characterize it. Sequence and structure constrain which molecular interactions are possible, but they do not determine whether the biological process producing a response occurs. The antigen must be processed into a short linear peptide and presented at sufficient abundance on at least one of the host’s MHC class I molecules. Cognate TCRs must be present in the

mature repertoire, and the resulting peptide–MHC–TCR interaction must occur under conditions that permit activation. We therefore treat immunogenicity as a context-dependent tendency for this biological process to produce a response, rather than as a fixed attribute of the molecular participants considered in isolation.

This context dependence changes how immunogenicity labels should be interpreted. When antigen supply, competitive presentation, repertoire selection, immune history, or exposure are absent from the modeled object, the same represented molecular configuration can correspond to different biological situations and different outcomes. Pooling observations while suppressing these distinctions collapses biologically different response-generating circumstances into the same modeled input. The resulting label variability is not necessarily unstructured noise. A peptide may remain below an activation threshold in one setting and cross it in another; a structurally compatible receptor may exist before selection but be absent from the mature repertoire; and the same peptide may occupy very different shares of presentation capacity in different HLA and peptide environments.

We therefore conjecture a mechanistic decomposition of the response-generating process. The model combines competitive peptide presentation, selection-conditioned TCR availability, and HLA-conditioned evidence of productive recognition. The decomposition does not establish that these components are complete or true. It makes explicit claims about how response is produced and separates choices that can be criticized at different levels.

We use *generalization* to mean the applicability of a conjectured regularity, mechanism, or model relation beyond the observations and biological circumstances that informed its construction. The object of a generalization claim may be a mechanistic relation, a component construction, a transferable parameter regime, an aggregation rule, or a composite model. Generalization is therefore claim- and context-relative and spans these levels. It is examined here through explicitly named changes in biological context. Estimating expected performance under repeated train–test sampling from a presumed population law is a separate inferential task.

We organize the empirical evidence as *refutation datasets*. A refutation dataset brings specified observations to bear on a conjecture and can eliminate a candidate when the declared criteria fail. The NCI tumor-infiltrating-lymphocyte neoepitope cohort resolves minimal epitopes and restricting HLA alleles. The PDAC cohort measures responses to personalized tumor-vaccine neoantigens. Two SARS-CoV-2 cohorts differ in antigen source and immune history: the spike-mutation cohort measures responses in vaccinated healthy donors, whereas the non-spike cohort measures responses in unvaccinated convalescent patients. These settings supply distinct opportunities for criticism without being treated as draws from a common population.

Their experimental resolution determines which model implications can be examined. Level 1 (L1) assigns a score to a minimal epitope paired with a restricting HLA allele. NCI observations are scored at this level. The PDAC and SARS-CoV-2 labels belong to long peptides containing several candidate minimal epitopes that may be presented by several HLA alleles. Level 2 (L2) therefore combines their candidate L1 scores into one score for the labeled long-peptide experimental unit.

Recognition geometry, a selection-decision regime, a component construction, and an L2 rule together specify a composite model. NCI can criticize its resolved-pair predictions but cannot distinguish models that differ only in L2. The long-peptide contexts can distinguish those models and can also expose deficiencies in their underlying L1 scores. This complementary resolution motivates

comparing complete composite models across all four datasets. Supported pairwise losses determine which models remain admissible in every context; the evidence need not identify a single survivor. Computational restrictions may require a sequential comparison, whose scope depends on which candidates and challengers it retains.

1 A mechanistic model of immunogenicity

CD8⁺ T-cell response to an antigen requires that the antigen is presented on class I MHC molecules, that there are available cognate TCRs, and that pMHC-TCR binding is sufficiently stable for the necessary downstream processes to complete. We define immunogenicity as the tendency for an antigen, specifically a minimal class I epitope e in an appropriate context, to elicit such a response. We quantify the post-immunological selection immunogenicity of an epitope–HLA pair (e, h) as

$$F(e, h, \mathcal{H}, \mathcal{S}; \theta) = P(e, \mathcal{H}, \mathcal{S}; \theta) \cdot N(e, \mathcal{H}, \mathcal{S}; \theta) \cdot Q(e, h, \mathcal{H}, \mathcal{S}) \cdot \Pi(e, h), \quad (1)$$

where Q quantifies pMHC abundance; $P \cdot N$ represents the availability of cognate TCRs capable of recognizing (e, h) following positive and negative (central and peripheral tolerance) selection; Π represents average pMHC–TCR interaction strength; $\mathcal{H}$ denotes the set of 6 class I MHC alleles associated with a particular person, with $h \in \mathcal{H}$; $\mathcal{S}$ denotes the self-peptidome.

The TCR-availability hypothesis space, Θ . Seven parameters govern the positive- and negative-selection factors. They divide into three biologically motivated groups. Recognition geometry is indexed by

$$g = (d_{\text{pos}}, \alpha_{\text{pos}}, d_{\text{neg}}, \alpha_{\text{neg}}).$$

Selection decisions are indexed by $d = (\mu, \eta)$. Selection exposure is indexed by τ_{expo} . Recognition geometry determines how cross-recognition decays with epitope–self-peptide distance. Selection exposure sets the effective scale of recognition opportunity during selection. Decision stringency sets the number of self-interactions required for positive selection or tolerated under negative selection pressure.

Let

$$\Theta = \mathcal{G} \times \mathcal{D} \times \mathcal{T},$$

where $\mathcal{G}$ is the set of recognition-geometry regimes, $\mathcal{D}$ is the set of selection-decision regimes, and $\mathcal{T}$ is the set of selection-exposure regimes. For $g \in \mathcal{G}$, $d \in \mathcal{D}$, and $\tau_{\text{expo},k} \in \mathcal{T}$, we write

$$\theta_{g,d,k} = (g, d, \tau_{\text{expo},k}).$$

The model evaluates the complete grid for each epitope–HLA observation. The resulting $F_{g,d,k}$ tensor records the response over recognition geometry, selection decisions, and exposure. Exposure remains a tensor coordinate, but is reduced by a fixed observation-level maximization within every geometry–decision regime before those regimes are compared.

TCR Availability. For each query epitope–HLA–person tuple, we compute the probability that post-selection cognate TCRs exist for (e, h) . This probability depends on how the query epitope relates to the self-peptide landscape in the given HLA context.

The biological intuition is clone-level. A generic TCR clonotype is more likely to pass positive selection when it has sufficient moderate self-recognition breadth in the thymus. We denote this conceptual breadth threshold by μ' . It summarizes the degree of moderate self-recognition associated with receiving survival-supporting signals; it is not a literal count of distinct binding events made by one physical TCR molecule.

The implemented model operates at the level of a latent conjugate set of TCR specificities capable of recognizing a query (e, h) , rather than enumerating individual clonotypes. The effective threshold μ asks whether that query-conditioned set has sufficient moderate cross-recognition support across $(\mathcal{S}, e; \mathcal{H})$. Similarly, η is an effective upper threshold on strong self cross-recognition, representing negative selection and tolerance pressure against strongly self-reactive cognate specificities.

We define Bernoulli random variables for each self peptide indicating whether it contributes an effective cross-recognition opportunity for the query-conditioned conjugate set. Summing these variables gives a hit count over distinct self peptides. We then compute the probability that this count is at least μ for positive selection and at most η for negative selection.

The self-peptidome contains millions of peptides. We compute hit-count distributions efficiently using either probability generating functions with the fast Fourier transform (FFT), or dynamic programming (DP).

Cross-recognition is defined formally as

$$p_{s,h}(e) = Q(s, h) \cdot \zeta(d[s, e]; d_{\text{threshold}}, \alpha)^2 \cdot \tau_{\text{expo}}. \quad (2)$$

This expression can be interpreted as the joint probability that the self peptide is presented (Q), that query-conjugate TCR specificities are structurally capable of cross-recognizing it given sequence similarity, and that this recognition opportunity is effectively sampled during selection. The factor $\zeta(d[s, e])$ captures the structural feasibility of cross-recognition given peptide similarity. It is a decreasing, scaled sigmoid that attains value one when s and e are identical and decays to zero as $d[s, e]$ increases.

Specifically,

$$\zeta(d[e, e']; d_{\text{threshold}}, \alpha) = \frac{1 + \exp(-d_{\text{threshold}} \alpha)}{1 + \exp((d[e, e'] - d_{\text{threshold}}) \alpha)}. \quad (3)$$

The midpoint $d_{\text{threshold}}$ and steepness α are set separately for positive ($d_{\text{threshold}}^{\text{pos}}, \alpha^{\text{pos}}$) and negative ($d_{\text{threshold}}^{\text{neg}}, \alpha^{\text{neg}}$) selection. See the Supplementary Information for a mathematical justification of the squared dependence on ζ and a discussion of the cross-reactivity distance function, $d[e, e']$.

The scalar τ_{expo} aggregates effective exposure to cross-recognition opportunities during selection, including encounter frequency and pre-selection TCR supply from VDJ recombination. These contributions enter the model as exogenous inputs. The available data do not directly identify the effective selection exposure for a query epitope–HLA–person tuple. We therefore evaluate a prespecified grid $\mathcal{T}$ and retain the largest modeled response over this coordinate separately for each observation and each candidate geometry–decision regime. The maximum treats effective selection

exposure existentially: a query receives the strongest cognate-TCR availability supported by any prespecified exposure regime, without identifying the regime that occurred.

The functions $P(e, \mathcal{H}, \mathcal{S}, \theta)$ and $N(e, \mathcal{H}, \mathcal{S}, \theta)$, computed from FFT or DP algorithms applied to the corresponding $p_{s,h}$ vectors, represent the probability of achieving sufficient moderate self cross-recognition breadth during thymic selection and avoiding excessive strong self cross-recognition, respectively. That is, they are the probabilities of passing positive and negative selection.

While thymic deletion is an important mechanistic contributor to this negative pressure, $N(e, \mathcal{H}, \mathcal{S})$ is intended to capture the aggregate action of central and peripheral tolerance mechanisms that suppress the action of, or eliminate, strongly self-reactive clones.

MHC Presentation. We define $Q(e, h, \mathcal{H}, \mathcal{S})$ as the relative abundance of the peptide–MHC complex (e, h) on the cell surface. Equivalently, Q measures the share of the class I presentation resource, across the HLA environment $\mathcal{H}$, allocated to presenting peptide e on allele h . This quantity is competitive: the presentation of e depends not only on its affinity for h , but also on the endogenous self-peptide background $\mathcal{S}$, the other alleles in $\mathcal{H}$, and the peptide supply available in the biological setting being modeled.

The model treats presentation as the outcome of four coupled influences. First, peptide supply determines how much peptide is available for loading. In *in vivo* settings, this supply depends on expression, gene or transcript abundance, and antigen processing. In assay settings, supply may instead be controlled by exogenous peptide dosing. Second, antigen processing affects how efficiently peptides enter the class I pathway. Third, binding affinity and pMHC stability determine how strongly and persistently a peptide can occupy a given HLA allele. Finally, all peptides compete for a finite pool of MHC molecules across the alleles in $\mathcal{H}$.

Thus $Q(e, h, \mathcal{H}, \mathcal{S})$ is a property of an epitope–HLA pair embedded in a specified competitive presentation context. Its value changes with the self-peptide background, peptide supply, HLA environment, and experimental or physiological setting. This context dependence matters when relating *in vitro* measurements to *in vivo* prediction: an assay can provide useful information about relative presentation or downstream immunogenicity while representing a different supply and competitive background from those operating in a person.

In practice, we compute Q using a closed-form competitive presentation approximation. The approximation combines effective peptide supply, antigen-processing scores, dissociation constants K_d , pMHC stability S , and reference MHC concentrations to estimate the fraction of cellular presentation capacity assigned to (e, h) .

For a set of peptides supplied in the modeled biological setting, let

$$\tilde{b}(e, h) = [M_{h,\text{ref}}] \frac{[P_e]_{\text{eff}} S(e, h)}{K_d(e, h)},$$

where $[P_e]_{\text{eff}}$ summarizes effective peptide supply, including abundance and processing, $S(e, h)$ is pMHC stability, and $K_d(e, h)$ is binding affinity. We then approximate presentation by

$$Q(e, h, \mathcal{H}, \mathcal{S}) = \frac{\tilde{b}(e, h)}{\sum_{h' \in \mathcal{H}} [M_{h',\text{ref}}] + \sum_{h' \in \mathcal{H}} \left(C_{h',\text{self}}^{\text{ref}}(\mathcal{S}) + \sum_{e' \in \mathcal{E}} \tilde{b}(e', h') \right)}.$$

Here, $\mathcal{E}$ denotes the set of non-self, experimentally supplied, or otherwise query peptides present in the modeled setting. The numerator is the effective presentation bid of peptide e on allele h : it increases with peptide supply and pMHC stability, and decreases as binding affinity weakens. The denominator normalizes this bid against the total class I presentation resource. It includes baseline MHC capacity, the competitive self-peptide background $C_{h',\text{self}}^{\text{ref}}(\mathcal{S})$, and the presentation bids of all supplied peptides across all alleles in $\mathcal{H}$.

This form makes $Q(e, h, \mathcal{H}, \mathcal{S})$ explicitly context dependent. Changing peptide supply, the set of competing peptides, the HLA environment, or the self-peptide background changes the fraction of presentation capacity assigned to (e, h) . In the vaccine setting, Q_{vax} is the same competitive presentation calculation evaluated under vaccine-induced antigen supply, with $\mathcal{E}$ set to the vaccine-derived peptide set and $[P_e]_{\text{eff}}$ reflecting vaccine antigen abundance over time. In non-vaccination settings, we treat $\mathcal{E}$ as empty.

Full details of the closed-form expression, context-specific background terms, and supply models are provided in the Supplementary Information.

pMHC–TCR Interaction Strength. The factor $\Pi(e, h)$ represents the residual pMHC–TCR interaction component of immunogenicity. It is intended to capture sequence-level evidence that a presented peptide–HLA complex has TCR-facing features associated with productive T-cell recognition, after MHC presentation and TCR availability have been represented by the other factors in the model.

The model contains no TCR-specific observation for the query tuple. We therefore estimate this component from HLA-conditioned similarity to two reference sets: experimentally confirmed immunogenic peptide–HLA pairs, $R_{\text{pos},h}$, and experimentally confirmed nonimmunogenic peptide–HLA pairs, $R_{\text{neg},h}$. These sets carry empirical information about latent biochemical features of pMHC–TCR interaction that remain unmodeled at the clone-specific level.

This component is deliberately predictive rather than causal. The reference peptide–HLA pairs do not cause the immunogenicity outcome of the query tuple. Their association with immunogenicity is therefore spurious in the causal sense, but it can still predict shared, latent TCR-facing biochemical features. $\Pi(e, h)$ exploits this noncausal correlation as limited evidence after the mechanistic presentation and TCR-availability pathways have been represented by Q , P , and N . It should not be interpreted as a causal mechanism linking the reference pairs to the query response.

The score is based on a gated density ratio,

$$\Pi(e, h) = \exp\left[w_{\min}(e)(\ln \rho_{\text{pos}}(e) - \ln \rho_{\text{neg}}(e))\right], \quad (4)$$

where

$$\begin{aligned} \rho_{\text{pos}}(e) &= \frac{1}{\sum_{r \in R_{\text{pos},h}} m_r} \sum_{r \in R_{\text{pos},h}} m_r \exp[-\beta_{\text{pos}} d(r, e)], \\ \rho_{\text{neg}}(e) &= \frac{1}{\sum_{r \in R_{\text{neg},h}} m_r} \sum_{r \in R_{\text{neg},h}} m_r \exp[-\beta_{\text{neg}} d(r, e)]. \end{aligned}$$

Here, m_r denotes the multiplicity of reference epitope r in the dataset, and $\beta_{\text{pos}} > 0$ and $\beta_{\text{neg}} > 0$ determine how sharply nearby reference points are weighted. When the query peptide lies near im-

munogenic examples relative to nonimmunogenic examples, the density ratio pushes $\Pi(e, h)$ above its neutral value of 1. When it lies near nonimmunogenic examples relative to immunogenic examples, the density ratio pushes $\Pi(e, h)$ below 1.

Since this factor intentionally uses noncausal reference-set associations, we include a safeguard against extrapolating those associations beyond the empirical support of the reference data. Let

$$d_+(e) = \min_{r \in R_{\text{pos},h}} d(r, e), \quad d_-(e) = \min_{r \in R_{\text{neg},h}} d(r, e).$$

We define

$$w_{\min}(e) = \exp \left[-\frac{1}{2} \lambda \min\{d_+(e), d_-(e)\}^2 \right].$$

The gate $w_{\min}(e)$ shrinks the log-density ratio toward zero when e is far from both reference sets. As $\min\{d_+, d_-\} \rightarrow \infty$, $w_{\min}(e) \rightarrow 0$ and $\Pi(e, h) \rightarrow 1$, so unsupported reference-set extrapolation is returned to the neutral value. Near regions covered by the reference data, $w_{\min}(e) \approx 1$, and $\Pi(e, h)$ recovers the full empirical density-ratio signal.

We also take care to reduce presentation-driven confounding in the construction of this term. Positive reference examples must have been sufficiently presented and must have supported productive T-cell recognition, whereas negative examples may be nonimmunogenic for several reasons, including poor presentation, limited TCR availability, or weak pMHC–TCR interaction. To bias the estimate toward TCR-facing signal, we estimate $(\beta_{\text{pos}}, \beta_{\text{neg}})$ using cross-validated conditional log-loss within binding-affinity strata. We then freeze these bandwidth parameters and optimize λ on a held-out set, allowing the density-ratio signal to contribute where reference support is strong while reducing overconfident predictions for peptides far from both reference manifolds. In practice, very few evaluated tuples receive any value other than 1 because, while the query epitope may appear close to an epitope in either reference set, it is rarely paired with the query HLA in either reference set.

Vaccine-conditioned extension. For revised PDAC scoring, vaccine-context presentation determines a bounded expansion term E_{vac} that modifies effective TCR availability:

$$F_{\text{vac}} = F(1 + E_{\text{vac}}).$$

This modification enters candidate epitope–HLA scoring before exposure maximization and long-peptide aggregation. Appendix D gives the construction and its parameterization. The vaccine parameters are fixed before candidate comparisons are formed.

Minimal-epitope scoring and long-peptide aggregation

The scoring procedure aligns mechanistic quantities with experimental units at two levels. Level 1 (L1) assigns a score to each minimal epitope–HLA pair under a specified recognition geometry, selection-decision regime, and component construction, maximizing over exposure separately for that pair. Level 2 (L2) combines these candidate-pair scores when the experimental label belongs to an unresolved long peptide. NCI scoring ends at L1. The external long-peptide cohorts require L2.

Exposure maximization within geometry and decision regimes. For every observation i and every candidate non-exposure parameter set (g, d) , we collapse that observation’s exposure axis by

$$\tilde{F}_i(g, d) = \max_{\tau_{\text{expo}, k} \in \mathcal{T}} F_i(g, d, \tau_{\text{expo}, k}). \quad (5)$$

This maximization is repeated within each regime, rather than choosing an exposure value once and reusing it across regimes. The maximizing grid value can differ between observations and between regimes for the same observation. The rule uses no outcome labels; ties are resolved by the lowest indexed exposure value.

Each combination of geometry–decision regime (g, d) and component construction c defines a candidate L1 scoring function:

$$S_i(g, d, c) = \max_{\tau_{\text{expo}, k} \in \mathcal{T}} F_{i,c}(g, d, \tau_{\text{expo}, k}). \quad (6)$$

Here $F_{i,c}$ is the tensor value under construction c , and the maximum is taken separately for every (i, g, d, c) . The exposure τ_{expo} is chosen by the fixed scoring rule within each observation and candidate regime, not learned as a shared parameter from the cohort labels. Each context uses its own biological inputs with the same specified geometry, decisions, exposure grid, and label-blind maximization. These candidate scores are defined before a metric is chosen to compare them. PR/CNAP and AUROC can criticize the same roster without requiring different parameter sets at the scoring stage.

The tournament compares the predictions produced by these candidate specifications. L1 and L2 define the scoring operations, whereas the tournament determines which combinations of parameter regime, component construction, and aggregation rule remain admissible.

Model component constructions. The component constructions specify the HLA environment used to compute Q , P , and N for each minimal epitope–HLA pair at L1. The full-HLA model uses the observed HLA environment for all three factors. The focal-HLA model retains that environment for Q but limits the self-pMHC roster entering P and N to the query HLA. The monoallelic model computes all three factors in a single-HLA environment. Two reciprocal constructions separate query presentation from selection-conditioned TCR availability: Mono-Q/full-PN combines monoallelic Q with full-HLA P and N , whereas Full-Q/mono-PN combines full-HLA Q with monoallelic P and N .

In all five constructions, $\Pi(e, h)$ depends on the query peptide and its restricting allele through the HLA-conditioned reference evidence. It has no dependence on the rest of the host’s HLA environment. Each construction is evaluated across the candidate recognition geometries and selection–decision regimes, producing the L1 scores supplied to L2 when the experimental unit is a long peptide. The component construction changes those input scores; the aggregation rule is specified separately.

Aggregation over unresolved experimental units. A new aggregation axis appears when the experimental label is attached to a long peptide rather than to a resolved minimal epitope–HLA pair. In that setting, the observed response may be caused by any of several candidate n -mer–HLA

tuples contained in the long peptide. For a long peptide ℓ , let

$$\mathcal{C}(\ell, \mathcal{H}) = \{(e_i, h) : e_i \subset \ell, h \in \mathcal{H}\}$$

denote the candidate presentation set. Write $S_{e,h}(g, d, c)$ for the L1 score of a candidate pair. Aggregation uses its numerical log score

$$z_{e,h} = \log[S_{e,h}(g, d, c) + \epsilon], \quad \epsilon = 10^{-12}.$$

The peptide and allele identities accompany each score because an aggregation rule may distinguish support from different peptides from support through different HLA alleles. A long-peptide aggregation functional A_ℓ therefore produces the L2 score from the indexed collection

$$S_{g,d,c,A_\ell}(\ell, \mathcal{H}, \mathcal{S}) = A_\ell(\{(e, h, z_{e,h}) : (e, h) \in \mathcal{C}(\ell, \mathcal{H})\}).$$

For each candidate model, the same aggregation function is used for every long peptide.

Seven alternative aggregation functions. We compare six direct score summaries with a seventh, hybrid aggregation rule. The six summaries are maximum, top-two mean, top-fraction mean, median, log-sum-exp, and log-mean-exp. They use the candidate scores without distinguishing peptide or allele identities. The seventh rule also uses those identities to separate support across distinct peptides from support through a second HLA allele. We call it a hybrid because it combines these two forms of support. All seven are alternative L2 functions, each producing one score from the same candidate roster; they are not successive steps of one scoring procedure.

The six direct summaries make different choices about how much of the score distribution to retain. Maximum uses only the strongest candidate. Top-two mean and top-fraction mean average the leading log scores, using the two highest-scoring candidates and the highest-scoring 2%, respectively. Median uses the center of the log-score distribution. Log-sum-exp instead accumulates positive-scale signal over all candidate pairs; log-mean-exp divides that accumulation by the number of pairs before taking its logarithm. The six summaries span reliance on a single leader, support among leading candidates, a typical candidate, and total or count-normalized accumulation. Appendix B gives their exact definitions, including the rounding conventions for small rosters.

The hybrid aggregation rule. Maximum preserves a strong candidate but ignores additional support. The hybrid retains that maximum m and adds two nonnegative increments. The peptide increment Δ_{pep} measures breadth across distinct short-peptide sequences, retaining only each sequence’s strongest HLA-specific score. An adaptive gate reduces the fraction of available breadth admitted as the resulting peptide score rises. The HLA increment Δ_{HLA} is a bounded bonus determined by the strength of the second-best distinct HLA route, which may involve the same peptide as the leading route or a different one. The final score is

$$S_{\text{hybrid}} = m + (1 - w)\Delta_{\text{pep}} + w\Delta_{\text{HLA}}, \quad 0 \leq w \leq 1.$$

The weight w controls the tradeoff between peptide and HLA support, while the maximum remains a lower bound. Appendix B defines both increments, their gates, and the calibration procedure.

The hybrid’s structure and constants were developed using full-HLA comparisons in PDAC, COVID spike, and COVID nonspike. Its constants are fixed across observations, geometry–decision regimes, component constructions, and both tournament metrics. Reusing the development contexts provides further criticism of the resulting model, not independent validation.

Experimental resolution. The experimental unit determines whether L2 is needed. NCI labels resolve a minimal epitope and restricting HLA, so its score ends at L1. Long-peptide assays require L2 after every candidate tuple has received an L1 score. Vaccine-context presentation and expansion terms enter each candidate score before this final aggregation.

For each geometry–decision regime, the five component constructions and seven specified L2 rules define 35 composite models. They share the specified scoring operations across contexts, although NCI does not use L2. The Results section develops the full comparison and the restrictions introduced by sequential approximations.

2 Refutation data

2.1 NCI neoepitope cohort

The NCI observations confront recognition-geometry and selection-decision regimes at the resolution of a minimal epitope and restricting HLA. Their L1 scores can distinguish component constructions and parameter regimes without introducing long-peptide aggregation. NCI can therefore supply an initial screen in a sequential tournament, while remaining one of the four contexts in the full comparison.

The NCI setting is derived from a tumor-infiltrating lymphocyte (TIL) neoepitope panel spanning 213 patients across 17 tumor histologies, with melanoma ($n = 66$) and colorectal cancers ($n = 56$ colon, $n = 16$ rectal, $n = 5$ colorectal) comprising the majority. Raw records were filtered to retain CD8⁺ T-cell responses only, and histologies were required to have an unambiguous TCGA expression category; 4,237 records were dropped at this stage. Each positive example is a minimal mutant (MT) 9-mer epitope whose immunogenicity was confirmed at the minimal epitope level by ex vivo TIL assay with verified CD8⁺ specificity. Each negative example is a set of contiguous 9-mers extracted from the wild-type (WT) long peptide corresponding to each MT epitope; protocol-confirmed assays established that TCRs cognate to the MT minimal epitope did not recognize the WT long peptide, and all 9-mer frames of the WT long peptide are therefore treated as non-immunogenic. This construction introduces a deliberate asymmetry: positive labels are confirmed at the minimal epitope level, whereas negative labels are confirmed at the long-peptide level, which is a coarser assay. The practical consequence is that the negative label excludes recognition of the WT long peptide by the MT-cognate TCR but does not rule out recognition of individual WT 9-mers by other TCRs in the patient repertoire; the negatives should therefore be understood as high-confidence rather than exhaustive. Each MT positive is assigned to the experimentally validated restricting allele (HLA-RE) only, and each WT 9-mer is similarly assigned to the HLA-RE of its paired MT row. Assigning WT peptides to all six patient alleles was explicitly rejected: a WT peptide with high affinity for an untested allele may be immunogenic on that allele, and including it would mislabel a potential positive as a confirmed negative. The assembled cohort contains 253

positive and 4,231 negative observations, with a positive class base rate of 5.6%. After requiring complete tensors, the retained cohort contains 250 positive and 4,180 negative observations.

The causal pairing between MT positives and WT negatives is preserved by construction: each pair shares patient, HLA allele, and tumor context. The tensor-complete observations supply the labels for supported comparisons among geometry–decision regimes and component constructions.

2.2 PDAC vaccine cohort

This refutation setting is derived from the phase I adjuvant mRNA neoantigen vaccine trial reported in Rojas et al. [3], in which 16 biomarker-evaluable patients with surgically resected pancreatic ductal adenocarcinoma (PDAC) received individualized autogene cevumeran followed by mFOLFIRINOX. Immunogenicity was assessed by ex vivo IFN γ ELISpot, which detects high-magnitude T-cell responses without distinguishing CD8 $^{+}$ from CD4 $^{+}$ activity. The supplementary spreadsheet contains 232 rows, each corresponding to one vaccine neoantigen defined by a long peptide of approximately 27 amino acids. Reproducing the analysis cohort in [3] requires four filtering decisions, each of which carries a causal assumption that bears on how discrimination tests from the resulting refutation setting should be interpreted.

The first decision removes two neoantigens with no usable ELISpot readout, leaving 230 neoantigens, which is the number the paper reports as administered across all patients. The second decision restricts the cohort to the eight patients who mounted at least one detectable ELISpot response. The stated justification is that a non-response in a non-responder is causally ambiguous: it could reflect genuine non-immunogenicity of the neoantigen or a failure of the patient’s immune system unrelated to neoantigen quality. Restricting to responders removes 124 neoantigens from non-responders and leaves 106. The third decision removes seven neoantigens from patient 25 that were tested in pools rather than individually, leaving 99 neoantigens with unambiguous individual labels. The fourth decision removes neoantigens for which no MHC-I-binding 9-mer subsequence containing the mutated residue is predicted at $K_d \leq 4,000$ nM, leaving the final cohort of 79 neoantigens of which 22 induced an ELISpot response. The neoantigen quality score is then defined as the mean over the top-two (9-mer, HLA) pairs by quality score, following [3].

Causal structure of the filtering chain. The four filtering decisions introduce three sources of systematic bias that bear on how discrimination tests from this refutation setting should be interpreted, and we represent these relationships explicitly in the directed acyclic graphs of Figure 1. Restricting the analysis to responders conditions on a common effect of neoantigen quality and patient immune fitness. Responder status is a collider: both neoantigen quality, which must be sufficient for at least one vaccine neoantigen to elicit a response, and immune fitness, which determines whether the patient’s immune system can mount any vaccine response at all, are upstream causes of whether a patient is classified as a responder. Conditioning on the collider opens a spurious association between neoantigen quality and immune fitness within the conditioned sample. Within responders, a neoantigen with low quality is systematically paired with higher immune fitness elsewhere in the patient than it would be in the unselected population, which biases the estimated association between quality score and immunogenicity toward underestimation. This collider bias is specific to the responder-restricted replication cohort. The expanded cohort defined below evalu-

ates every long peptide, including those from non-responders. Responder status therefore remains unconditioned, the collider path stays closed, and this source of bias is removed.

In the replication cohort, the $K_d \leq 4,000$ nM filter introduces selection bias on a causal mediator. Binding affinity to MHCI is causally upstream of both the quality score and immunogenicity; dropping neoantigens without a qualifying binder therefore removes whole units from the low end of the score distribution. This is mediator selection: binding affinity is a component of the quality score and a causal precursor of immunogenicity. The removed units therefore carry structurally low scores; among them are immunogenic neoantigens the class I model necessarily scores low, such as the ELISpot-positive neoantigen lost in this filter (discussed below). Removing units the model would rank low but that are in fact positive inflates the reported discrimination relative to the unfiltered candidate pool. The expanded cohort removes this selection mechanism at its source. It retains every long peptide and assigns the floor $\log\text{-}F$ value when all candidate epitopes fail the presentation filters. The filter therefore acts within a retained observation’s score while the evaluated population remains fixed. A long peptide with surviving candidates is scored from those survivors; a long peptide with none is retained at the floor.

The restriction to 9-mers in [3] follows necessarily from their cross-reactivity distance metric, which is defined on 9-mer pairs via an independent site approximation and is formally undefined for other lengths. This constraint compounds the mediator selection bias described above: neoantigens whose immunogenic minimal epitope is a 10-mer or 11-mer are either dropped entirely or scored on a suboptimal 9-mer subsequence. Because HLA allele determines which peptide lengths are preferentially presented, the resulting measurement error in the quality score is systematic rather than random, correlated with the patient HLA profile. We extend scoring to 9–12-mers by assigning each non-9-mer epitope the minimum cross-reactivity distance over its constituent 9-mer subsequences, decoupling the distance metric definition from the scoring domain. This is an approximation: the minimum 9-mer distance is a lower bound on the true distance between a longer epitope and any reference, which is conservative in the sense of overestimating cross-reactivity. Epitopes shorter than nine residues and longer than twelve are excluded for practical reasons unrelated to the distance metric. The empirical question of whether this approximation introduces more error than the coverage restriction removes is addressed directly by the expanded cohort analysis described below.

The ELISpot outcome measure introduces a third distortion whose resolution requires lineage-resolved outcome data. The assay detects IFN γ production from both CD8 $^+$ and CD4 $^+$ T-cells without distinguishing between them, whereas the quality score addresses only the MHCI-restricted CD8 $^+$ pathway. A neoantigen labeled immunogenic in the ELISpot may owe its label to CD4 $^+$ activity rather than to any MHCI-presented epitope, as is directly illustrated by the one ELISpot-positive neoantigen lost in the K_d filter: it produced a detectable response despite no predicted MHCI binder, implying a CD4 $^+$ -mediated or very weakly presented CD8 $^+$ response. The practical consequence is that the outcome label and the model score are causally connected through different pathways for different examples in the dataset, introducing label noise whose direction and magnitude require additional lineage-resolved data to determine.

In the filtered replication cohort the three biases act in different directions. Collider bias attenuates discrimination; mediator selection bias inflates it; lineage confounding adds noise of indeterminate sign. The net effect on the reported AUC of 0.68 in [3] is therefore ambiguous, and any single AUC-

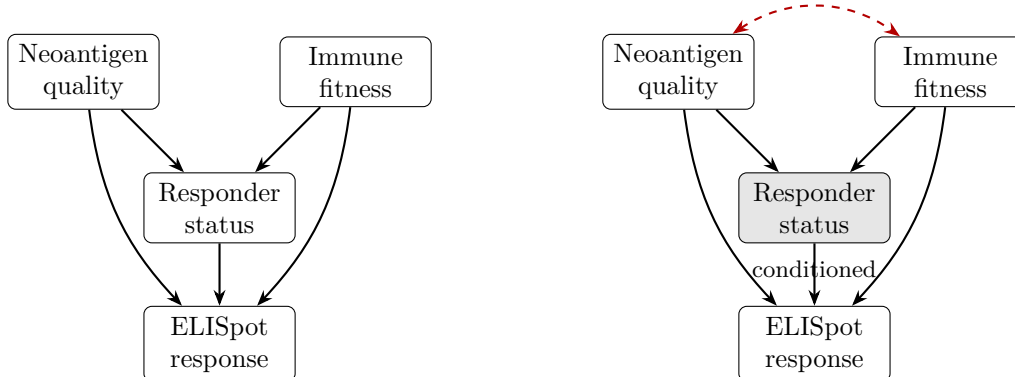


Figure 1: Collider bias induced by restricting analysis to responders. Left: the uncontrolled DAG, in which neoantigen quality and immune fitness are independent causes of responder status and of ELISpot response. Right: conditioning on responder status (shaded) opens a spurious bidirected path between neoantigen quality and immune fitness (dashed red arc), biasing the estimated quality–immunogenicity association toward underestimation. The expanded cohort corresponds to the left (uncontrolled) DAG: evaluating every long peptide, including those from non-responders, leaves responder status unconditioned and the quality–fitness path closed. The right panel depicts the bias induced by the responder-restricted replication cohort.

based discrimination figure derived from the filtered cohort should be understood as a distorted reflection of the model’s true discrimination on the clinical candidate population. The expanded cohort removes the two unit-selection biases: it retains non-responders, closing the collider, and retains filtered long peptides at the floor $\log\text{-}F$ value, so mediator selection no longer acts on the evaluated population. Lineage confounding from the $\text{CD8}^+/\text{CD4}^+$ ELISpot readout remains and requires lineage-resolved data. We therefore weight the expanded-cohort results toward the shared substrate claim rather than the finer selection-regime contrast.

Original and context-revised PDAC refutations. The original tensor-complete expanded cohort contains 213 long-peptide endpoints, including 23 positive and 190 negative observations. This original PDAC refutation retains every eligible patient and uses the baseline score $F = P \cdot N \cdot Q \cdot \Pi$. It asks whether the composite models survive criticism in the full represented PDAC context, treating patient immune fitness as contextual heterogeneity outside the antigen-specific score.

A clinically important consideration is that several trial participants had undergone splenectomy. Splenectomy may alter the capacity to mount or sustain the measured immune response through a patient-level pathway outside antigen-specific F . Removing splenectomy-positive patients yields a context-revised collection of 191 endpoints, including 19 positive and 172 negative observations. For that collection we will replace the baseline score with F_{vac} (Appendix D), which explicitly represents vaccine-conditioned presentation and TCR expansion, and repeat the same directed round robin.

This planned comparison contrasts the original end-to-end composite models with versions revised for the PDAC context. Two features change together: the revised cohort excludes splenectomy-positive patients, and the revised score adds vaccine-conditioned presentation and TCR expansion. A difference between the two graphs therefore describes the consequence of the combined revision. The present comparison asks whether that revised biological account changes the non-dominated

survivor set. Assigning an observed change to one feature would require two additional comparisons: splenectomy exclusion while retaining the baseline score, and vaccine-conditioned scoring while retaining the full cohort. Because the current revisions are confronted with the same PDAC labels, this is an adaptive continuation of conjecture and criticism. Future PDAC observations will provide new opportunities to refute the revised survivors.

2.3 SARS-CoV-2 spike-mutation cohort

The third refutation setting is derived from Bozkus et al. [4] (Fig.2E), which profiled T-cell responses to spike mutations in variants of concern in a cohort of 12 healthy donors who had received either mRNA-1273 or BNT162b2 vaccination encoding the Wuhan-1 spike sequence. Blood was collected 14 days after the second dose (V2D14). T-cells were expanded by stimulation with pooled 15mer overlapping peptides spanning spike mutations found in Alpha, Beta, and Gamma variants, then re-stimulated with 24 individual 15mer peptides — each spanning a single mutation site in either the Wuhan-1 (WT) or variant (MT) sequence — and IFN- γ production by CD4⁺ and CD8⁺ T-cell subsets was measured by intracellular cytokine staining.

Each observation is a (peptide, donor) pair with a continuous %IFN- γ ⁺ readout. The setting contains two conditions: WT→WT, in which T-cells expanded on the Wuhan-1 pool are re-stimulated with individual Wuhan-1 peptides, and MT→MT, in which T-cells expanded on the variant pool are re-stimulated with individual variant peptides. Full six-locus HLA typing is available for each donor.

The readout values are DMSO-adjusted %IFN- γ ⁺ frequencies, with donor-level background subtraction applied prior to analysis. A (peptide, donor) observation is binarized as positive if the background-subtracted frequency exceeds a minimum detection floor $\epsilon > 0$:

$$Y_{ij} = \begin{cases} 1 & \text{if } x_{ij} > \epsilon \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

where x_{ij} is the background-subtracted %IFN- γ ⁺ frequency for peptide i in donor j , and ϵ is a minimum detection floor applied to remove residual technical noise near zero. The cohort spans 23 mutation sites and 12 donors, producing at most 276 observations per condition per T-cell subset.

Causal structure and limitations. Several features of the data generating process bear on how discrimination tests from this refutation setting should be interpreted.

All donors were previously vaccinated against Wuhan-1 spike prior to the expansion assay. This affects both conditions. The WT→WT condition is the more directly affected: the vaccine encodes Wuhan-1 spike, so the expanded T-cell population is enriched for vaccine-primed memory cells rather than naive T-cells primed in vitro. The MT→MT condition is similarly affected, because vaccine-primed memory T-cells that cross-recognize variant peptides contribute to the expansion and re-stimulation signal. Both conditions therefore measure vaccine-conditioned recall and cross-recognition. Relative to a naive-repertoire assay of de novo immunogenicity, this design compresses the dynamic range between high- and low-scoring predictions.

The experimental unit is an unresolved donor–15mer response. Each mutation site is covered by

multiple overlapping 15mers at 5 amino acid offsets, and the re-stimulation uses individual 15mers from the pool rather than a single peptide per site. A positive readout therefore reflects recognition of at least one minimal epitope contained within the responding 15mer, on at least one of the donor’s HLA alleles. The causal minimal epitope and restricting allele remain latent. Minimal epitopes may be 9–12-mers, and each 15mer contains multiple overlapping frames of each length. Model scoring therefore requires an aggregation strategy over the set of (mer, HLA) pairs induced by the donor’s six-locus HLA type, of which taking the maximum or the mean of the top- k scores are special cases.

The readout is dominated by $CD4^+$ responses: anti-spike T-cells are predominantly $CD4^+$ in vaccinated donors [4], and the model targets the MHC class I-restricted $CD8^+$ pathway. $CD8^+$ signal is present but sparse, and point estimates of $CD8^+$ -specific discrimination will be sensitive to the small number of positive (peptide, donor) pairs in that subset.

2.4 SARS-CoV-2 non-spike cohort

The fourth refutation setting is derived from Bozkus et al. [4] (Fig. 4A), which deconvolved T-cell responses to individual non-spike SARS-CoV-2 epitopes in a cohort of 13 unvaccinated convalescent patients (ATLAS cohort). T-cells were expanded by stimulation with pooled 15mer overlapping peptides spanning selected regions of non-spike proteins, including the entire nucleocapsid (N) and selected regions of membrane (M), envelope (E), ORF3a, ORF6, and ORF7a/b. Expanded T-cells were then re-stimulated with the individual 15mer peptides constituting those pools, and polyfunctional cytokine production ($IFN-\gamma^+/TNF-\alpha^+$) by $CD4^+$ and $CD8^+$ T-cell subsets was measured by intracellular cytokine staining. Full six-locus HLA typing is available for each patient. Each observation is a (peptide, patient) pair with a continuous $\%IFN-\gamma^+/TNF-\alpha^+$ readout reported after donor-level DMSO background subtraction. Per-patient binary labels are derived by applying the same detection-floor binarization defined in Equation 7, yielding a positive label if the specific patient’s background-subtracted response exceeds ϵ and a negative label otherwise. Combined with per-patient HLA typing, this produces (peptide, patient, HLA, label) tuples directly usable for model scoring.

Relationship to the spike-mutation cohort. This cohort complements the spike-mutation cohort in several respects. The donors are unvaccinated convalescent patients rather than vaccine recipients, removing the dominant Wuhan-1 spike memory priming that compresses dynamic range in the Fig. 2E data. The peptide source is also broader, spanning selected regions of multiple non-spike proteins rather than 23 mutation sites within spike alone, which tests generalization across a wider sequence and protein space. Together these differences mean the two SARS-CoV-2 refutation settings probe the model under distinct biological conditions while sharing the same fundamental refutation task: binary immunogenicity classification at the (peptide, donor) level using the binarization defined in Equation 7.

Causal structure and limitations. Several features of the data generating process bear on how discrimination tests from this refutation setting should be interpreted.

The expansion pools cover selected regions of non-spike proteins rather than their complete sequences. Detection is therefore conditional on priming by those selected regions. Responses to peptides outside the pools remain unobserved, leaving the corresponding false-negative rate unidentified from these data. Per-patient positivity should consequently be interpreted as recognition conditional on pool-level priming.

As with the spike cohort, the experimental unit remains unresolved at the minimal-epitope level, and $CD8^+$ responses are again sparser than $CD4^+$ responses.

3 Results

3.1 Composite models span resolved and unresolved experimental units

A recognition geometry and selection-decision regime, evaluated under one component construction, give an L1 score for each minimal epitope–HLA observation. Each score is the maximum of that observation’s F tensor over exposure for the specified regime and construction (Equation (6)). These scores directly confront the NCI labels. For a long peptide, an L2 rule combines the candidate minimal-epitope–HLA scores into one prediction that can confront its response label. A complete composite model therefore specifies

$$M = (g, d, c, A_\ell), \tag{8}$$

where g is the recognition geometry, d the selection-decision regime, c the component construction, and A_ℓ the aggregation rule. Varying these choices produces the full roster $\mathcal{V}$ of competing models, each with predictions in all four contexts.

Two models that differ only in their aggregation rule make identical NCI predictions. The long-peptide datasets can distinguish them because they confront the aggregated scores. Those datasets also criticize the underlying L1 choices: changing the geometry, decisions, or component construction changes the scores being aggregated. The full tournament allows all four datasets to bear on these complete models before any alternatives are removed through operational preference in a single context.

3.2 Each biological context can exclude a composite model

Suppose two models, A and B, rank the same PDAC observations differently. An empirical advantage for A becomes a supported claim to replace B only if it survives the declared challenges. PR/CNAP examines whether the advantage persists across target prevalences; AUROC examines its dependence on the weights assigned to represented positive and negative cases. Paired computational challenges examine its dependence on combinations of the observed units. Supported magnitude and literal survival summarize these comparisons under the policy specified in Appendix C.

A supported loss in PDAC excludes B from a claim of admissibility throughout all four contexts. B’s performance elsewhere cannot cancel that loss, and A need not also replace B in NCI or either SARS-CoV-2 cohort. Each context can expose a deficiency, possibly through a different rival. Applying this rule to every model pair leaves a survivor set in each context. The joint survivors

are the models common to all four sets:

$$S_0 = S_{\text{NCI}} \cap S_{\text{PDAC}} \cap S_{\text{spike}} \cap S_{\text{nonspike}}. \quad (9)$$

Here S_e contains the models with no supported incoming loss in context e under the current policy. The graph formulation and the conditions justifying this exclusion rule are given in Appendix C.

The intersection may contain several models or none. Several survivors mean that the specified comparisons have not excluded those alternatives; an empty set means that no model in the roster survives every context. If practical use requires choosing among several survivors, operational rules can express a preference without changing which models passed the criticism policy. The survivor set S_0 is therefore retained separately from the operational selection. PR/CNAP and AUROC produce separate sets over the same model roster; neither assigns a probability to performance in an unrealized cohort.

3.3 Computational limits of the full tournament

Crossing 4,200 geometry–decision regimes with five component constructions and seven L2 rules produces 147,000 composite models. Comparing every pair in all four contexts would require approximately 4.32×10^{10} pair–context matches per metric, with both directions assessed in each match. Deciding which models survive need not require all of these comparisons.

Once a model has a supported loss, further losses are unnecessary to establish its exclusion. Establishing survival is more demanding: every rival that could supply an incoming loss must still be accounted for. The accelerated implementation uses this asymmetry. It rules out directions that cannot pass the observed-metric gate and stops assessing a target within a context once a supported loss has been found. It retains the evidence for that exclusion. For a reported survivor, every possible incoming loss must have been assessed or ruled out by the observed-metric gate. An assessed comparison may remain scientifically unresolved; such a comparison supplies no supported loss. Excluded models remain available as rivals in comparisons with other models.

This procedure preserves the joint survivor set, but its cost depends on how many comparisons are needed to establish losses and survival. Worst-case work remains quadratic in the number of models. If many models survive, a subsequent operational selection can also be expensive: the current refinement requires complete numerical comparisons among those survivors across contexts. These costs motivate restricting the roster before constructing and comparing all long-peptide models.

3.4 NCI survivors define reduced composite-model rosters

NCI can supply the first comparison because its observations require only L1 scores. Within each component construction, the NCI tournament compares the candidate geometry–decision regimes and retains every regime without a supported incoming loss. This survivor set is recorded before the later operational rules select one or more preferred regimes.

Each surviving regime can then be paired with all seven L2 rules. For example, a surviving full-HLA regime produces seven complete models with the same underlying minimal-epitope scores but

different rules for combining those scores into a long-peptide prediction. Repeating this construction for every NCI survivor across the five component constructions produces a reduced roster for comparison in NCI, PDAC, spike, and non-spike. Appendix A gives its formal definition.

NCI remains part of this second tournament because the initial tournaments compare regimes only within each component construction. The second tournament also compares models from different constructions and can therefore expose NCI losses that the initial comparisons did not examine. The long-peptide contexts additionally distinguish the aggregation rules and the consequences of the underlying L1 scores.

This sequence preserves every regime that survived its initial NCI tournament. Whether the resulting roster is tractable depends on how many regimes survive. A smaller roster could be formed from the NCI operational selections or from representatives of distinct surviving regimes, but those restrictions would omit alternatives that NCI had not excluded through a supported loss.

3.5 Omitted models can still supply decisive criticism

A model can lose its eligibility to be selected while remaining useful for criticizing another model. Suppose model A has a supported loss in NCI, but supports replacement of model B in PDAC. The full tournament excludes A because of its NCI loss and B because of its PDAC loss. A reduced tournament that removes A before the PDAC comparisons may leave B’s deficiency unexposed.

A model retained for its ability to supply such a comparison is a challenger, even if it is no longer eligible to be selected. Preserving these challenger roles separates exact staged execution from the NCI survivor approximation. An NCI loss excludes every aggregation variant of the losing regime because those variants share the same NCI predictions. The variants can cease to be eligible winners, but they must remain available to challenge other models where required by the full comparison. If the second tournament instead compares only models constructed from NCI survivors, it can miss losses supplied by excluded challengers.

Passing only NCI operational selections introduces another omission. A regime that survived supported criticism but lost the NCI operational comparison might survive the full joint tournament or become its preferred choice after another context excludes the NCI-preferred alternative. Choosing representatives of surviving regimes can omit such alternatives as well. Even identical NCI rankings do not establish that two regimes make identical predictions in the long-peptide contexts.

These omissions can be examined by expanding the restricted comparison. Restoring excluded models as challengers tests whether they supply additional supported losses. Restoring omitted survivors as eligible candidates tests whether they remain admissible or change the operational selection. With the scoring rules, observations, and criticism policy held fixed, each change can be attributed to the added comparisons. Agreement between restricted rosters still leaves the unexamined comparisons open. The choice of approximation therefore depends on the number and diversity of the NCI survivors and on which omitted comparisons can subsequently be restored.

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A NCI screening and operational selection

For each component construction, the NCI roster contains 4,200 geometry–decision regimes. Every observation is scored by the same maximum over its six exposure values within each regime. The complete roster includes steepness variants and regimes with identical score vectors; empirical AP and AUROC do not screen their inclusion. Separate PR/CNAP and AUROC tournaments apply the supported-replacement policy to these scores.

The fixed-policy graph reduction produces $S_{0,\text{NCI},c}$, the surviving geometry–decision regimes for component construction c . Let $\mathcal{C}$ contain the five component constructions and $\mathcal{A}$ the seven aggregation rules. The full composite-model roster is $\mathcal{V} = \mathcal{G} \times \mathcal{D} \times \mathcal{C} \times \mathcal{A}$. Expanding every component-specific NCI survivor into all aggregation variants gives

$$\mathcal{V}_{\text{NCI}} = \bigcup_{c \in \mathcal{C}} \{(g, d, c, A_\ell) : (g, d) \in S_{0,\text{NCI},c}, A_\ell \in \mathcal{A}\}. \quad (10)$$

These initial survivor sets are metric-specific. They omit cross-component NCI comparisons and can therefore retain regimes that a pooled NCI tournament would exclude.

When an operational selection is requested, every member of $S_{0,\text{NCI},c}$ remains an eligible candidate and challenger. For each candidate, incoming activation strengths are sorted from greatest to least. Their vectors are minimized lexicographically, comparing the strongest threat first and proceeding to the next coordinate only on a tie. The resulting set T may remain plural.

Only then is metric-specific regret compared, retaining all of $S_{0,\text{NCI},c}$ as benchmarks. PR uses worst-target CNAP regret over the declared prevalence interval. AUROC uses empirical AUROC regret at uniform within-class weights. Identical final vectors remain tied in S_{op} . For the current single-context NCI AUROC policy, empirical AUROC maximizers survive the observed gate and have zero incoming activation profiles; the regret rule therefore retains those maximizers. This property does not extend to a joint comparison in which another context can exclude them.

Expanding $S_{0,\text{NCI},c}$ or expanding S_{op} defines different reduced composite-model rosters. The former preserves every regime that passed the fixed NCI criticism policy, whereas the latter also restricts eligibility through the NCI operational rule. Neither restricted roster alone preserves the full joint tournament’s challenger obligations.

B Long-peptide aggregation systems

Scope and common inputs. A long-peptide assay supplies one label without identifying which candidate short-peptide–HLA pair explains the response. Level 2 must map these candidate-level predictions to one score for that experimental unit. We specify seven alternative aggregation functions: maximum, top-two mean, top-fraction mean, median, log-sum-exp, log-mean-exp, and a hybrid aggregation rule. The first six directly summarize the candidate scores. The seventh also uses peptide and allele identities, combining distinct-peptide breadth with support through a second HLA allele; this combination is what we call the hybrid. The functions are competing choices for L2, not successive stages of the calculation. This appendix defines the common inputs and the six direct summaries before developing the hybrid.

For a fixed geometry–decision regime and component construction, let $S_{jh} \geq 0$ be the candidate L1 score and write $z_{jh} = \log(S_{jh} + \epsilon)$, with $\epsilon = 10^{-12}$. Here j indexes a distinct short-peptide sequence and h its candidate restricting allele. Each peptide–HLA tuple is counted once, giving K candidate pairs. All seven functions receive this same indexed collection of log scores.

All seven rules are endpoint-local: they use only the endpoint’s candidate scores and identities and constants fixed before scoring. Adding an unrelated long peptide to a batch therefore cannot change an existing endpoint’s score. Each rule is applied unchanged across the five component constructions and all geometry–decision regimes; it leaves the underlying L1 component calculations unchanged.

Six direct score summaries. For these operators, flatten the candidate roster to $z_1, \dots, z_K$ and order the values as $z_{(1)} \geq \dots \geq z_{(K)}$. Define $k_2 = \min(2, K)$ and $k_f = \max(1, \lceil 0.02K \rceil)$ for a nonempty roster. The definitions and the information retained by each summary are:

Function	Definition	Information retained
Maximum	$z_{(1)}$	The strongest candidate alone
Top-two mean	$k_2^{-1} \sum_{i=1}^{k_2} z_{(i)}$	The mean of the leading two log scores
Top-fraction mean	$k_f^{-1} \sum_{i=1}^{k_f} z_{(i)}$	The mean of the leading 2% of log scores
Median	$\text{median}(z_1, \dots, z_K)$	The center of the log-score distribution
Log-sum-exp	$\log\left(\sum_{i=1}^K e^{z_i}\right)$	Accumulated positive-scale signal over all pairs
Log-mean-exp	$\log\left(K^{-1} \sum_{i=1}^K e^{z_i}\right)$	Positive-scale accumulation normalized by pair count

The two upper-tail means average log scores; log-mean-exp averages their exponentials before taking the logarithm. These are different operations. All six functions depend only on the multiset of scores: reassigning those scores among peptide and allele identities leaves their outputs unchanged. Consequently, they do not distinguish whether additional high scores belong to different peptides or to one peptide paired with several alleles.

Why add a hybrid alternative? Maximum protects a strong candidate from dilution by numerous weak alternatives, but discards all secondary support. The other direct summaries retain additional scores without distinguishing their peptide and allele identities. The hybrid combines a maximum anchor with two separate summaries of additional support. It makes three commitments: preserve the strongest candidate’s score, allow additional candidate support to raise it, and distinguish support across peptide sequences from support through different HLA alleles. These are modeling hypotheses, not properties uniquely determined by the assay. The six direct summaries remain alternatives against which those choices can be assessed.

Two summaries of the peptide-by-HLA matrix. For the hybrid, view the same roster as a matrix whose rows are distinct short-peptide sequences j and whose columns are distinct candidate restricting alleles h . Retaining the maximum along each row or column answers a different question:

$$\underbrace{u_j = \max_h z_{jh}}_{\text{best score for peptide } j}, \quad \underbrace{b_h = \max_j z_{jh}}_{\text{best score for allele } h}.$$

Both summaries retain the same overall maximum, $m = \max_j u_j = \max_h b_h$, but discard different information. The following configurations illustrate the distinction; “strong” describes candidate model scores, not experimentally confirmed epitopes or calibrated response probabilities.

Candidate-score pattern	Peptide-level support	HLA-level support
One strong peptide–HLA pair	One leading peptide	One leading allele
Several strong peptides, one allele	Additional peptides	One leading allele
One strong peptide, several alleles	One leading peptide	Additional alleles
Several strong peptides, several alleles	Additional peptides	Additional alleles

Row maxima give each peptide one contribution, preventing its scores under several alleles from being counted as several distinct peptides. Column maxima recover information about alternative alleles that row maxima may discard. Neither distinct sequences nor distinct alleles imply independent biological evidence: candidate peptides can overlap, and the two summaries come from the same roster.

The score as a maximum plus two increments. Let Δ_{pep} denote the admitted peptide-support increment and Δ_{HLA} the bounded second-allele increment, defined below. The final score is

$$S_{\text{hybrid}} = m + (1 - w)\Delta_{\text{pep}} + w\Delta_{\text{HLA}}, \quad 0 \leq w \leq 1. \quad (11)$$

This form separates the protected maximum from the two forms of corroboration. With $s_{\text{pep}} = m + \Delta_{\text{pep}}$ and $s_{\text{HLA}} = m + \Delta_{\text{HLA}}$, it is equivalently the convex combination $(1 - w)s_{\text{pep}} + ws_{\text{HLA}}$. Convex weighting is an additional modeling choice that controls the tradeoff between the increments. Increasing w raises the coefficient of HLA support and lowers that of peptide support; it does not append an HLA bonus to an unchanged peptide score. Retaining the shared maximum once does not by itself require this particular weighting.

The available peptide-support increment. On the row-collapsed roster, let $x_j = \max_h S_{jh}$ be each peptide’s positive-scale signal before addition of the numerical floor. For a nonzero roster, define

$$T = \sum_j x_j, \quad M = \max_j x_j, \quad R = \frac{T}{M}, \quad p_j = \frac{x_j}{T}.$$

The ratio R measures total signal relative to the strongest peptide. It equals one if only one peptide carries positive signal, and equals n if exactly n peptides carry equal positive signal. More generally it is the order-infinity Hill effective number, $N_\infty = 1/\max_j p_j = R$, rather than a literal count of peptides.

The collapsed log-sum score $U = \log(T + \epsilon)$ exceeds the maximum $m = \log(M + \epsilon)$ by an available accumulation $U - m$. Temper this accumulation by the fraction of signal outside the leader:

$$D = 1 - \max_j p_j = 1 - \frac{1}{R},$$

$$C_{\text{pep}} = D \max(U - m, 0) = \left(1 - \frac{1}{R}\right) \log\left(\frac{T + \epsilon}{M + \epsilon}\right).$$

The maximum with zero is a numerical safeguard, since $T \geq M$. When M is large relative to the numerical floor,

$$C_{\text{pep}} \approx \left(1 - \frac{1}{R}\right) \log R.$$

Thus the two factors depend on the same effective-breadth ratio; they are not independently varying descriptions of the roster. Their product suppresses small amounts of nonleading signal more strongly than the untempered accumulation: if $R = 1 + r$ with small positive r , then, away from the floor, $C_{\text{pep}} \approx r^2$, whereas $U - m \approx r$.

This summary has a specific limitation. Rosters with the same maximum and total receive the same offer regardless of how their remaining signal is divided. For example, positive-scale rosters $(1, 1)$ and $(1, \frac{1}{2}, \frac{1}{2})$ have the same M , T , and C_{pep} , despite containing different numbers of nonzero peptides. The offer captures dominance-based effective breadth after sequence collapse, not the full distribution of peptide support.

Admitting peptide support through a self-limiting gate. The available offer does not specify how much should influence the final score. We conjecture that peptide corroboration should contribute less as the resulting score becomes strong. A decreasing sigmoid implements this choice smoothly; the assay does not require either that conjecture or this particular functional form. Write the admitted increment as

$$\Delta_{\text{pep}} = C_{\text{pep}} \sigma\left(\frac{c_{\text{pep}} - m - \Delta_{\text{pep}}}{\kappa_{\text{pep}}}\right), \quad \sigma(v) = \frac{1}{1 + e^{-v}}, \quad (12)$$

with $s_{\text{pep}} = m + \Delta_{\text{pep}}$ and $\kappa_{\text{pep}} > 0$. The center c_{pep} is the resulting score at which half the offer is admitted; the width κ_{pep} controls the smoothness of that transition.

The offer raises the score, while that rise reduces the fraction admitted. This negative feedback explains why the increment appears on both sides of Equation 12. Gating only on m would fix the fraction before the increase occurred. Here the left side increases with Δ_{pep} and the right side is nonincreasing; their intersection exists uniquely in $[0, C_{\text{pep}}]$.

The sharp-transition limit makes the behavior explicit. Holding all other quantities fixed gives

$$s_{\text{pep}} \xrightarrow{\kappa_{\text{pep}} \rightarrow 0^+} m + \min\{C_{\text{pep}}, (c_{\text{pep}} - m)_+\}, \quad (a)_+ = \max(a, 0).$$

In this limit, a maximum below the center receives the full offer if the resulting score would remain below the center; otherwise it rises only to the center. A maximum already above the center is preserved. For positive transition width the behavior is smooth, and the center is not a strict cap.

Qualifying support through a second HLA allele. The allele maxima b_h retain information about additional alleles even when the same short peptide supplies their strongest scores. Let $b_{(2)}$ be the second-largest maximum among distinct alleles. Its own strength qualifies the additional route: a small gap from the leader would also reward two similarly weak scores. An increasing sigmoid admits a fraction of a nonnegative bonus cap B :

$$\Delta_{\text{HLA}} = B\sigma\left(\frac{b_{(2)} - t_{\text{HLA}}}{\delta_{\text{HLA}}}\right), \quad s_{\text{HLA}} = m + \Delta_{\text{HLA}}, \quad (13)$$

where $\delta_{\text{HLA}} > 0$. The bonus is $B/2$ when $b_{(2)} = t_{\text{HLA}}$, and the positive width controls the transition. The second allele’s strongest candidate may be the same peptide as the leading allele’s candidate or a different one.

The gates answer different questions. The decreasing peptide gate regulates how much an available accumulation may raise the resulting score; the increasing HLA gate assesses whether a second allele’s input score warrants a bonus. Adding that bonus does not change $b_{(2)}$, so the HLA gate requires no self-consistency equation. Both sigmoids are deterministic scoring operations, not response probabilities or established biological thresholds.

Bounds, limiting cases, and score scale. The increments satisfy $0 \leq \Delta_{\text{pep}} \leq C_{\text{pep}}$ and $0 \leq \Delta_{\text{HLA}} \leq B$. Consequently,

$$m \leq S_{\text{hybrid}} \leq m + (1 - w)C_{\text{pep}} + wB.$$

Neither branch dilutes the strongest candidate, and the HLA contribution cannot exceed wB . A single positive peptide gives no peptide offer, but can still receive HLA corroboration if it has support through a second allele. Conversely, a roster with several supported peptides but only one represented allele can supply peptide breadth without an HLA increment. With $w = 0$, the score is the peptide branch; with $w = 1$, it is the HLA branch.

Relative breadth and absolute score location play different roles. Multiplying every positive-scale candidate score by a common positive factor leaves R unchanged. Away from the numerical floor it also leaves C_{pep} unchanged, but shifts the log scores entering the gates. The gate centers therefore refer to the specified L1 score scale. With fixed centers, the admitted increments can change under rescaling even when relative breadth does not.

Numerical conventions. The numerical floor is $\epsilon = 10^{-12}$. In computation, recover the positive-scale peptide signal from its log score by $x_j = \max(e^{u_j} - \epsilon, 0)$ before forming T , M , and p_j . Otherwise the floor would itself accumulate as apparent support. An empty roster, or a roster with every $x_j = 0$, returns $\log \epsilon$ before normalization. When fewer than two distinct alleles are represented, set $\Delta_{\text{HLA}} = 0$. Bisection solves the peptide equation within $[m, m + C_{\text{pep}}]$ for s_{pep} , to an absolute bracket width of 10^{-10} , with at most 64 iterations.

Calibration of the hybrid. The functional form leaves the gate locations, widths, bonus cap, and mixture weight unspecified. Calibration used completed full-HLA candidate scores in PDAC, COVID spike, and COVID nonspike. Candidate structures and parameterizations were compared with the same full-HLA fixed-maximum reference within each context. The staged paired-CNAP assessment used 600 computational replications and required a positive supported magnitude with survival above zero exceeding 0.81 in every context. Eligibility was therefore conjunctive across contexts. The comparisons informed successive structural and numerical revisions, rather than selecting the final hybrid from a single grid declared before development.

The frozen specification is

$$(c_{\text{pep}}, \kappa_{\text{pep}}, t_{\text{HLA}}, \delta_{\text{HLA}}, B, w) = (-2.2, 0.13, -6.45, 0.02, 1, 0.12).$$

It admits half the peptide offer at $s_{\text{pep}} = -2.2$ and half the HLA bonus at $b_{(2)} = -6.45$. The mixture assigns weights 0.88 and 0.12 to the peptide and HLA branches, respectively, limiting the final HLA increment to 0.12 log-score units. A local sensitivity assessment held the peptide gate fixed and evaluated the Cartesian product

$$\begin{aligned} t_{\text{HLA}} &\in \{-6.475, -6.45, -6.425\}, & \delta_{\text{HLA}} &\in \{0.015, 0.020, 0.025\}, \\ B &\in \{0.50, 0.75, 1.00\}, & w &\in \{0.08, 0.10, 0.12\}. \end{aligned}$$

Each of these 81 parameterizations was assessed by the same paired criterion in all three contexts to probe sensitivity to the numerical choices. The frozen specification above is used unchanged for both tournament metrics and every component construction. Response labels informed its development, but neither labels nor context identity enter its endpoint-scoring function.

The seven aggregation rules and five component constructions define 35 composite models for each geometry–decision regime. Crossing these models with 4,200 regimes gives the 147,000-model roster used for both metrics. Each NCI survivor contributes seven aggregation variants to a reduced roster.

C Supported-superiority tournament policy

Each pair of composite models supplies two directed replacement claims within each of the four contexts. The proposed comparison retains the NCI policy: supported magnitude must exceed zero, and literal survival above zero must exceed 0.81 under order-two computational challenges formed from 600 paired replications. PR uses prior-standardized average precision over target prevalences from 1% to 30%, expressed as $\text{CNAP}_\pi = (\text{AP}_\pi - \pi)/(1 - \pi)$. AUROC challenges the paired advantage by varying within-class case weights. The observed anchor and computational assessment must both meet the declared requirements; neither metric uses an exchangeable-label reference law.

Every context supplies a separate assessment. The comparison preserves model pairing on the same experimental units, and its computational construction must specify which units are resampled together to preserve the dependencies admitted by that context. Repeated observations and linked patient or peptide records require an explicit unit specification. Neither pooling all cohort rows nor averaging cohort advantages implements the candidate-conservative rule.

A context’s supported directions form a graph G_e , with an edge from the supported replacement to the model it exceeds. A strongly connected component contains models joined by directed paths in both directions. A source component has no incoming edge from another component. The models in these source components form the context’s maximal set, $S_e = \text{srcSCC}(G_e)$. Candidate-conservative selection intersects these sets:

$$S_0 = \bigcap_{e \in \mathcal{E}} \text{srcSCC}(G_e), \quad \mathcal{E} = \{\text{NCI, PDAC, spike, nonspike}\}.$$

The strict observed CNAP and AUROC gates provide increasing scalar potentials for supported directions under the current policy, so each context graph is acyclic. Its source components are individual models with no incoming supported loss, as used in Equation (9). One incoming edge therefore supplies an exclusion witness.

The alternative replacement-conservative policy first intersects the directed edge sets and then finds the source components of the resulting graph. It requires the same replacement direction to be supported in every context. Different rivals can therefore expose a model’s deficiencies in different contexts without excluding it from this alternative selection. The candidate-conservative rule instead allows each context to exclude a model through its own supported comparisons.

Accelerated candidate-conservative execution certifies exclusions and resolves survivor obligations against the full declared roster. An excluded model remains a challenger. Reuse of identical scores or paired computations must preserve the scientific policy, computational design, seeds, and audit bindings; shared predictions alone do not authorize substituting an existing match artifact. The accelerated output certifies selection and need not contain a complete graph or the alternative replacement-conservative reduction.

Operational refinement freezes the joint S_0 as both candidate and challenger set. Incoming vectors combine activation strengths across contexts and challengers and are minimized lexicographically. If several candidates remain in T , PR compares worst-target CNAP regrets against every member of S_0 ; AUROC compares empirical AUROC regrets in each context. Descending context-regret vectors are minimized lexicographically, and exact ties remain ties. Complete numerical reports among survivors are required for this refinement even when fewer reports suffice to certify S_0 .

A sequential approximation must record its stage-specific candidate and challenger sets. Removing a model from eligibility after a supported NCI loss can preserve the joint selection; removing its unassessed challenger roles can omit exclusions elsewhere. A restriction based on operational preference or representative selection can additionally omit eligible candidates. Changing those restrictions, the spike observation definition, or the PDAC scoring and inclusion rules defines a revised comparison whose conclusions must retain that provenance.

D Vaccine-conditioned extension of the immunogenicity score

In a vaccine setting, the immune system is exposed to a coordinated set of vaccine-derived peptides at high effective concentration. This creates an additional layer of structure. Vaccine-derived n-mers compete with one another for presentation, and the resulting pMHC complexes can expand the TCR clones available to recognize related peptide–HLA contexts.

Let $\mathcal{V}$ denote the set of vaccine-derived n-mers induced by the administered long peptides. For each $e \in \mathcal{V}$, we define a vaccine-context binding bid

$$b_{\text{vac}}(e, h) = \frac{[P_e]_{\text{vac}} S(e, h)}{K_d(e, h)}, \quad \tilde{b}_{\text{vac}}(e, h) = [M_{h, \text{tot}}] b_{\text{vac}}(e, h),$$

where $[P_e]_{\text{vac}}$ is the effective vaccine-derived peptide supply. The corresponding vaccine presentation measure is

$$Q_{\text{vac}}(e, h, \mathcal{H}, \mathcal{V}) = \frac{\tilde{b}_{\text{vac}}(e, h)}{\sum_{h' \in \mathcal{H}} [M_{h', \text{ref}}] + \sum_{h' \in \mathcal{H}} \left(C_{h', \text{self}}^{\text{ref}}(\gamma) + \sum_{e' \in \mathcal{V}} \tilde{b}_{\text{vac}}(e', h') \right)}.$$

This expression uses the same calibrated-background normalization as Q , with the query set replaced by the full vaccine-derived peptide set. High vaccine supply increases the total vaccine bid mass, which is then divided competitively among vaccine-derived n-mers. Each n-mer receives a share of the available HLA-specific presentation capacity according to the strength of its binding bid relative to endogenous background and to the bids of the other vaccine-derived n-mers. Relative presentation therefore remains selective even under strong vaccine loading.

We next define the vaccine-induced stimulation load for a scored observation $o = (e, h, \mathcal{H})$:

$$A_{\text{vac}}(o) = \sum_{v \in \mathcal{V}} K_{\text{TCR}}(o, v) Q_{\text{vac}}(v).$$

Here $K_{\text{TCR}}(o, v)$ encodes sharing of TCR recognition between the scored observation o and the vaccine-derived pMHC v . The simplest choice is an identity kernel, where only the same n-mer–HLA pair contributes. More general kernels can assign partial weight to related pMHCs that are expected to share cross-reactive TCR clones.

We convert this stimulation load into a bounded expansion term,

$$E_{\text{vac}}(o) = \beta \frac{A_{\text{vac}}(o)^\alpha}{K_A^\alpha + A_{\text{vac}}(o)^\alpha},$$

where β is the maximal additive vaccine-induced expansion term, so that the largest possible multiplicative increase in effective TCR availability is $1 + \beta$. K_A is the stimulation level at which expansion reaches half of its maximal value, and α controls the sharpness of the transition.

The vaccine-conditioned score scales effective TCR availability $P \cdot N$ by $1 + E_{\text{vac}}$:

$$F_{\text{vac}}(e, h, \mathcal{H}, \mathcal{S}; \theta) = P(e, \mathcal{H}, \mathcal{S}; \theta) \cdot N(e, \mathcal{H}, \mathcal{S}; \theta) \cdot Q(e, h, \mathcal{H}, \mathcal{S}) \cdot (1 + E_{\text{vac}}(e, h, \mathcal{H}, \mathcal{V})) \cdot \Pi(e, h). \quad (14)$$

In the initial implementation, we use the identity form of K_{TCR} and treat β , K_A , and α as global calibration parameters. The values used in a vaccine-conditioned tournament specify the vaccine-conditioned terms of the revised composite models and are recorded before their pairwise comparisons are formed. Extensions with cross-reactive kernels or exhaustion terms can be introduced once the data support those additional degrees of freedom.

Scale of the vaccine expansion parameters. The parameters in E_{vac} should be interpreted on the scale of the stimulation load A_{vac} , not on the scale of administered vaccine dose. In the identity-kernel implementation, $A_{\text{vac}}(e, h)$ is simply the vaccine-context presentation share $Q_{\text{vac}}(e, h, \mathcal{H}, \mathcal{V})$ for the same peptide–HLA pair. Thus, K_A represents the per-observation vaccine presentation level required for half-maximal expansion. A value such as $K_A = 1$ would place half-maximal expansion at the extreme where essentially all vaccine-context presentation mass is concentrated in a single peptide–HLA observation. Competition among many vaccine-derived n-mers places the empirical A_{vac} distribution well below that scale. We therefore set or fit K_A relative to the empirical distribution of A_{vac} values. For example, choosing K_A near the upper tail of this distribution makes expansion selective for the most strongly presented vaccine pMHCs, whereas a smaller K_A allows moderate presentation levels to induce appreciable expansion. The parameter β controls the maximal possible additive increase in effective TCR availability, while α controls how sharply expansion transitions around K_A .

In the default implementation, we use $\rho_{\text{vac}} = 2$, $\beta = 1$, $K_A = 10^{-2}$, and $\alpha = 1$. The choice $\rho_{\text{vac}} = 2$ sets the total vaccine-derived bid mass to twice the calibrated endogenous self-background bid mass, representing a strong but not dominant vaccine presentation pressure. The choice $\beta = 1$ allows vaccine exposure to at most double the effective TCR availability term through the factor $1 + E_{\text{vac}}$. The value $K_A = 10^{-2}$ means that a peptide–HLA observation reaching approximately one percent vaccine-context presentation share attains half-maximal expansion. Finally, $\alpha = 1$ gives a smooth Michaelis–Menten-like transition rather than an abrupt threshold. These defaults serve as interpretable starting values whose biological adequacy is examined through calibration and sensitivity analysis.

Status of the vaccine-conditioned score. The baseline score $F = P \cdot N \cdot Q \cdot \Pi$ and the vaccine-conditioned score F_{vac} define different PDAC predictions. A comparison must specify which version supplies the PDAC context and fix its vaccine parameters before assessing the resulting composite models. Replacing the baseline predictions with vaccine-conditioned predictions is a revision of that context’s scoring rule. Combining this change with splenectomy exclusion also changes the admitted observations, so attribution requires the separate comparisons described in the PDAC refutation data. These revisions continue conjecture and criticism against the existing PDAC evidence.

Bid-level parameterization of vaccine peptide supply Administered RNA or peptide dose provides only indirect information about the intracellular supply of individual vaccine-derived n-mers. Processing, expression, and degradation intervene between administered dose and the peptides available for HLA loading. In parallel, the endogenous background term $C_{h,\text{self}}^{\text{ref}}(\gamma)$ is a calibrated effective competition scale. We therefore parameterize vaccine supply at the level most relevant to the presentation model: the relative bid mass of vaccine-derived peptides compared with the endogenous self background.

Let

$$B_{\text{self}}(\mathcal{H}, \gamma) = \sum_{h \in \mathcal{H}} C_{h,\text{self}}^{\text{ref}}(\gamma)$$

denote the total calibrated self-background bid scale for HLA environment $\mathcal{H}$ and context class γ . We introduce a dimensionless implementation parameter ρ_{vac} , defined so that the total vaccine-

derived bid mass satisfies

$$\sum_{h \in \mathcal{H}} \sum_{e \in \mathcal{V}} \tilde{b}_{\text{vac}}(e, h) = \rho_{\text{vac}} B_{\text{self}}(\mathcal{H}, \gamma).$$

Thus, $\rho_{\text{vac}} = 1$ corresponds to vaccine-derived competitive pressure comparable to the calibrated endogenous self background, while larger or smaller values represent stronger or weaker vaccine loading, respectively.

To convert ρ_{vac} into peptide-level supplies, we assign each vaccine-derived n-mer a nonnegative relative supply weight u_e . In the simplest implementation, $u_e = 1$ for all $e \in \mathcal{V}$. We then set

$$[P_e]_{\text{vac}} = c_{\text{vac}} u_e,$$

where the scale factor c_{vac} is chosen to enforce the desired total vaccine bid mass:

$$c_{\text{vac}} = \frac{\rho_{\text{vac}} B_{\text{self}}(\mathcal{H}, \gamma)}{\sum_{h \in \mathcal{H}} \sum_{e \in \mathcal{V}} [M_{h, \text{tot}}] u_e S(e, h) / K_d(e, h)}.$$

This yields

$$\tilde{b}_{\text{vac}}(e, h) = [M_{h, \text{tot}}] \frac{c_{\text{vac}} u_e S(e, h)}{K_d(e, h)}.$$

Under this parameterization, $[P_e]_{\text{vac}}$ is an effective coordinate on a relative presentation-competition scale. The resulting bid preserves the model's dependence on peptide–HLA binding strength, stability, HLA environment, and the endogenous calibrated background.